Category

Yield-Based Bond Convexity and Portfolio Properties

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BOND CONVEXITY AND CONVEXITY ADJUSTMENT

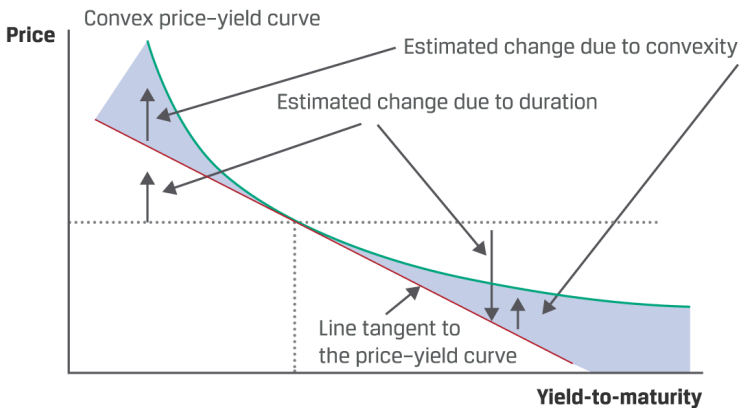
Learning Outcome

- calculate and interpret convexity and describe the convexity adjustment

Interest rate risk, or the sensitivity of a bond's price to changes in yield, is an important concept used to analyze fixed-income exposures. As a key risk metric, modified duration measures the first-order (linear) effect on a bond's price change given a yield-to-maturity change. **Convexity** is a complementary risk metric that measures the second-order (non-linear) effect of yield changes on price for an option-free fixed-rate bond, as shown in Exhibit 1.

Exhibit 1:

Convexity of an Option-Free Fixed-Rate Bond



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The true relationship between a bond's price and its yield-to-maturity is the curved (convex) line that shows the actual bond price given its market discount rate. Duration (i.e., money duration) estimates the change in the bond price along the straight line tangent to the curved line. For small yield-to-maturity changes, there is little difference between the lines.

However, for larger yield changes, the difference becomes significant and duration alone is an insufficient risk measure. In such cases, a proper assessment requires capturing the non-linear price effect of convexity. By considering duration and convexity metrics together for relatively large yield changes, the estimated bond prices are substantially closer to the actual bond prices. That is, when including convexity, bond prices rise more when yields decrease and fall less when yields increase than when estimated using duration alone.

As shown in Equation 1, convexity adds to the estimate of the percentage (full) price change provided when using modified duration by itself, which was used in prior lessons.

$$\%\Delta PV^{Full} \approx (-AnnModDur \times \Delta Yield) + \left[\frac{1}{2} \times AnnConvexity \times (\Delta Yield)^2\right]. \qquad 1$$

The first expression in parentheses is the effect from modified duration. The expression in brackets is the **convexity adjustment**: the annual convexity statistic, *AnnConvexity*, times one-half times the change in the yield-to-maturity *squared*. This term is always positive for an option-free fixed-rate bond, so, as noted, the bond price is higher for either an increase or decrease in yield (see Exhibit 1).

1 and 2.

Consider once again BRWA's five-year, 3.2% (semiannual) coupon bond priced at par for settlement on 15 October 2025 and maturity on 15 October 2030. Exhibit 2 is an interactive spreadsheet in the downloadable Microsoft Excel workbook that shows the calculation of *AnnConvexity* for this bond; notice that it is very similar to the one used to calculate duration. The difference is the inclusion of a new column (Col. 7) for the following operation (where *CF* means cash flow) to calculate convexity:

(Time to receipt of *CF*) × (Time to receipt of *CF* + 1) × (Weight of *CF*) × (1 + Periodic *YTM*)$^{(-\text{Periods per year})}$.

Note that the intermediate result of multiplying the first two terms introduces non-linearity (Time to receipt of CF^2). Dividing the sum of Column 7 (96.95578) by periods per year squared (2^2=4) results in an annualized convexity (*AnnConvexity*) statistic of 24.2389.

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Exhibit 2:

Convexity Calculation for an Option-Free Fixed-Rate Bond

BRWA Corporation Five-Year, 3.2% Semiannual Bond at Issuance	
Principal	$300,000,000
Fixed Coupon (%)	3.20
Periods p.a.	2
Price (per 100 Par Value)	100
Yield-to-Maturity (%)	3.20

Col. 1 Period	Col. 2 Time to Receipt	Col. 3 Cash Flow	Col. 4 Present Value	Col. 5 Weight	Col. 6 = Col. 2 × Col. 5 Time to Receipt × Weight	Col. 7 = Col. 2 × (Col. 2 + 1) × Col. 5 × (1 + YTM/2)$^{-2}$ Convexity of Cash Flows
1	1.0	1.6	1.5748	0.0157	0.0157	0.0305
2	2.0	1.6	1.5500	0.0155	0.0310	0.0901
3	3.0	1.6	1.5256	0.0153	0.0458	0.1774
4	4.0	1.6	1.5016	0.0150	0.0601	0.2909
5	5.0	1.6	1.4779	0.0148	0.0739	0.4295
6	6.0	1.6	1.4546	0.0145	0.0873	0.5919
7	7.0	1.6	1.4317	0.0143	0.1002	0.7767
8	8.0	1.6	1.4092	0.0141	0.1127	0.9829
9	9.0	1.6	1.3870	0.0139	0.1248	1.2093
10	10.0	101.6	86.6875	0.8669	8.6688	92.3766
			100.0000	**1.0000**	**9.3203**	**96.9558**
Annualized Macaulay Duration and Convexity					**4.6601**	**24.2389**

Exhibit 2 can also be viewed as a spreadsheet on the candidate learning ecosystem online in the downloadable Microsoft Excel workbook.

Alternatively, Equation 2 can be used to approximate annualized convexity (*ApproxCon*). This approach is useful for bonds with uncertain cash flows, such as those with contingency features and default risk, which will be explored in later lessons. Note that this equation uses the same inputs as approximating modified duration.

$$ApproxCon = \frac{(PV_-)+(PV_+)-[2\times(PV_0)]}{(\Delta Yield)^2\times(PV_0)}. \qquad 2$$

PRICE(*settlement, maturity, rate, yield, redemption, frequency, [basis]*),

we have the following:

PV_+ = PRICE(DATE(2025,10,15),DATE(2030,10,15),0.032,0.0325,100,2,0) = 99.770965.

PV_- = PRICE(DATE(2025,10,15),DATE(2030,10,15),0.032,0.0315,100,2,0) = 100.229641.

Applying Equation 2, *ApproxCon* is estimated to be 24.23896, which is a very close approximation to the bond's *AnnConvexity* of 24.23895 we calculated in Exhibit 2.

$$ApproxCon = \frac{100.229641 + 99.770965 - (2 \times 100.00)}{(0.0005)^2 \times 100.00} = 24.23896.$$

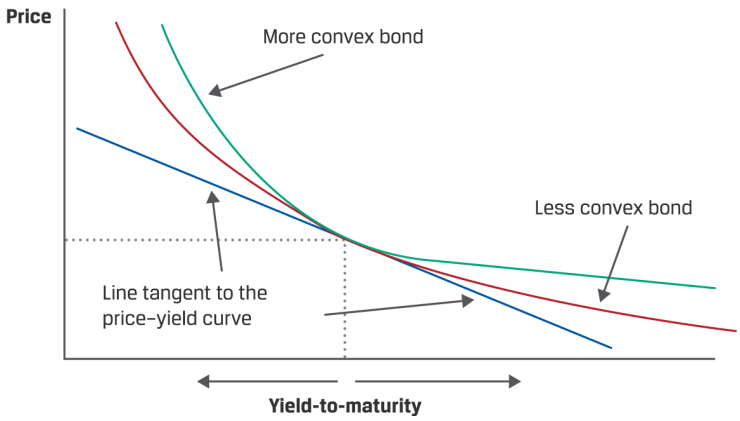
Bond features that lead to greater convexity are the same as for duration. A fixed-rate bond will have greater convexity:

- the *longer* its time-to-maturity,
- the *lower* its coupon rate, and
- the *lower* its yield-to-maturity.

Another factor is the dispersion of cash flows, meaning the degree to which payments are spread out over time. For two bonds with the same duration, the one with the greater dispersion of cash flows has greater convexity. The positive attributes of convexity for an investor are shown in Exhibit 3.

Exhibit 3:

Positive Attributes of Bond Convexity on an Option-Free Bond



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The two bonds depicted in Exhibit 3 are assumed to have the same price, maturity, yield-to-maturity, and modified duration, so they share the same line tangent to their price–yield curves. The benefit of greater convexity arises when their yields change significantly. For the same decrease in yield-to-maturity, the more convex bond *appreciates more* in price, while for the same increase in yield-to-maturity, the more convex bond *depreciates less* in price. Consequently, the more convex bond outperforms the less convex bond in both falling and rising yield environments, so it is less risky for investors.

This conclusion assumes the difference in convexity is not reflected in the price of the more convex bond. If it is priced in, the more convex bond would have a higher price (and lower yield-to-maturity). The key point is that convexity is valuable to investors for risk reduction, so they must pay for it.

Duration and Convexity of 30-Year Romanian Bond

A 30-year bond issued by the government of Romania has a coupon rate of 4.625% (paid annually) and currently trades at a yield-to-maturity of 4.75% on 15 October 2025, with a maturity date of 15 October 2055.

1. Use Excel's PRICE function to calculate *ApproxModDur* and *ApproxCon* using a 5 bp increase and decrease in yield-to-maturity.
2. Compare and interpret the duration and convexity measures of the 30-year Romanian bond and BRWA's 5-year bond.

To find *ApproxModDur* and *ApproxCon*, we must first calculate PV_0 and then, using a 5 bp increase and decrease in YTM, calculate PV_+ and PV_-:

PV_0 = PRICE(DATE(2025,10,15),DATE(2055,10,15),0.04625,0.0475,100,1,0) = 98.022448.

PV_+ = PRICE(DATE(2025,10,15),DATE(2055,10,15),0.04625,0.0480,100,1,0) = 97.247386.

PV_- = PRICE(DATE(2025,10,15),DATE(2055,10,15),0.04625,0.0470,100,1,0) = 98.806567.

$$ApproxModDur = \frac{98.806567 - 97.247386}{2 \times (0.0005) \times 98.022448} = 15.90637.$$

$$ApproxCon = \frac{98.806567 + 97.247386 - (2 \times 98.022448)}{(0.0005)^2 \times 98.022448} = 369.58881.$$

Duration and Convexity Comparison: Romanian and BRWA Bonds

Bond	Maturity (Years)	Coupon (%)	YTM (%)	*Approx-ModDur*	*ApproxCon*
Romanian	30	4.625 Annual Pay	4.75	15.90637	369.64203
BRWA	5	3.20 Semi-Annual	3.20	4.58676	24.23896

The duration and convexity of the Romanian bond are significantly higher than BRWA's bond primarily due to its much longer time-to-maturity (30 years versus 5 years), which outweighs the Romanian bond's higher coupon rate and YTM, features that decrease duration and convexity.

Question Set

Questions 1–5 relate to the following information.

Bond	Issuance Date	Maturity	Annual Pay Coupon (%)	YTM (%)
Bond X	7/15/25	7/15/34	2.25	2.25
Bond Y	7/15/25	7/15/35	5.33	2.34

1. Calculate the modified duration and convexity for Bond X at issuance.

Hide Solution

Solution:

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Price (per 100 Par Value)	100
Yield-to-Maturity (%)	2.25

Col. 1	Col. 2	Col. 3	Col. 4	Col. 5	Col. 6 = Col. 2 × Col. 5	Col. 7 = Col. 2 × (Col. 2 + 1) × Col. 5 × (1 + YTM)$^{-1}$
Period	**Time to Receipt**	**Cash Flow**	**Present Value**	**Weight**	**Time to Receipt × Weight**	**Convexity of Cash Flows**
1	1.0	2.25	2.200489	0.02200	0.02200	0.04304
2	2.0	2.25	2.152067	0.02152	0.04304	0.12628
3	3.0	2.25	2.104711	0.02105	0.06314	0.24701
4	4.0	2.25	2.058398	0.02058	0.08234	0.40262
5	5.0	2.25	2.013103	0.02013	0.10066	0.59064
6	6.0	2.25	1.968805	0.01969	0.11813	0.80870
7	7.0	2.25	1.925481	0.01925	0.13478	1.05454
8	8.0	2.25	1.883111	0.01883	0.15065	1.32601
9	9.0	102.25	83.693835	0.83694	7.53245	73.66694
			100.0000	**1.00000**	**8.24718**	**78.26579**
Annualized Macaulay Duration and Convexity					**8.24718**	**78.26579**
Modified Duration					**8.06571**	**—**

2. Calculate *ApproxModDur* and *ApproxCon* for Bond Y using a 5 bp increase and decrease in the yield-to-maturity.

Hide Solution

Solution:

To find *ApproxModDur* and *ApproxCon* for Bond Y, we must first calculate PV_0, PV_+, and PV_-.

PV_0 = PRICE(DATE(2025,7,15),DATE(2035,7,15),0.0533,0.0234,100,1,0) = 126.386358.

$PV+$ = PRICE(DATE(2025,7,15),DATE(2035,7,15),0.0533,0.0239,100,1,0) = 125.877943.

$PV_$ = PRICE(DATE(2025,7,15),DATE(2035,7,15),0.0533,0.0229,100,1,0) = 126.897325.

$$ApproxModDur = \frac{126.897325 - 125.877943}{2 \times (0.0005) \times 126.386358} = 8.06560.$$

$$ApproxCon = \frac{126.897325 + 125.877943 - (2 \times 126.386358)}{(0.0005)^2 \times 126.386358} = 80.76704.$$

3. Compare and interpret the risk measures from Questions 1 and 2 relative to a small anticipated change in yield-to-maturity.

Hide Solution

Solution:

For a small change in YTM, modified duration is a reasonable estimate of the price change risk of a bond. Since the modified duration (or *ApproxModDur*) values of Bond X and Bond Y are virtually identical (8.06571 and 8.06560, respectively), the estimated price change based on a small change in YTM (either positive or negative) would be nearly identical for each bond.

Solution:

For a large change in YTM, duration alone does not capture the expected price change of a bond. The convexity adjustment accounts for the non-linear nature of the relationship of price to YTM. Since both bonds have virtually the same duration, the higher convexity of Bond Y should result in a smaller percentage drop given a large increase in YTM and a larger percentage increase given a large decrease in YTM. In either case, Bond Y should perform better than Bond X for a large change in YTM.

5. Compare and interpret each of the following features for its effect on the convexity of Bond X versus Bond Y:

a. Maturity

b. Coupon rate

c. YTM

Hide Solution

Solution:

a. Bond Y has a longer maturity than Bond X (10 years versus 9 years), which would increase Bond Y's convexity relative to Bond X.

b. Bond X has a lower coupon rate compared to Bond Y (2.25% versus 5.33%), which would increase Bond X's convexity relative to Bond Y.

c. Bond X has a lower YTM compared to Bond Y (2.25% versus 2.34%), which would increase Bond X's convexity relative to Bond Y.

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SA

Does anyone understand why a lower coupon rate leads to higher convexity? Conceptually it makes sense: the principal payment makes up a larger part of the total return of the bond and therefore the lower coupon bond is more exposed to interest rate risk (correct?). However, can we prove this mathematically?

Created 6 months ago by Sophie Anne Kamphuis — 3 replies | Last Activity: 2 months ago

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MP

anyone knows how tto calculate moddur and convexity in BA II Plus (basic version). It drives me crazy that even through the same example i end up having different results

Created 6 months ago by Manuel Paredes Caldito — 4 replies | Last Activity: 3 months ago

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SD

In Question 1 is this formula correct? Col. 7 = Col. 2 × (Col. 2 + 1) × Col. 5 × (1 + YTM)^-1

Shouldn't the ending be x (1 + YTM)^-2?

Created 5 months ago by Sherwyn Dominic Marc Harris — 1 reply | Last Activity: 5 months ago

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